



AquaHyel

Water Hyacinth Removal Robot

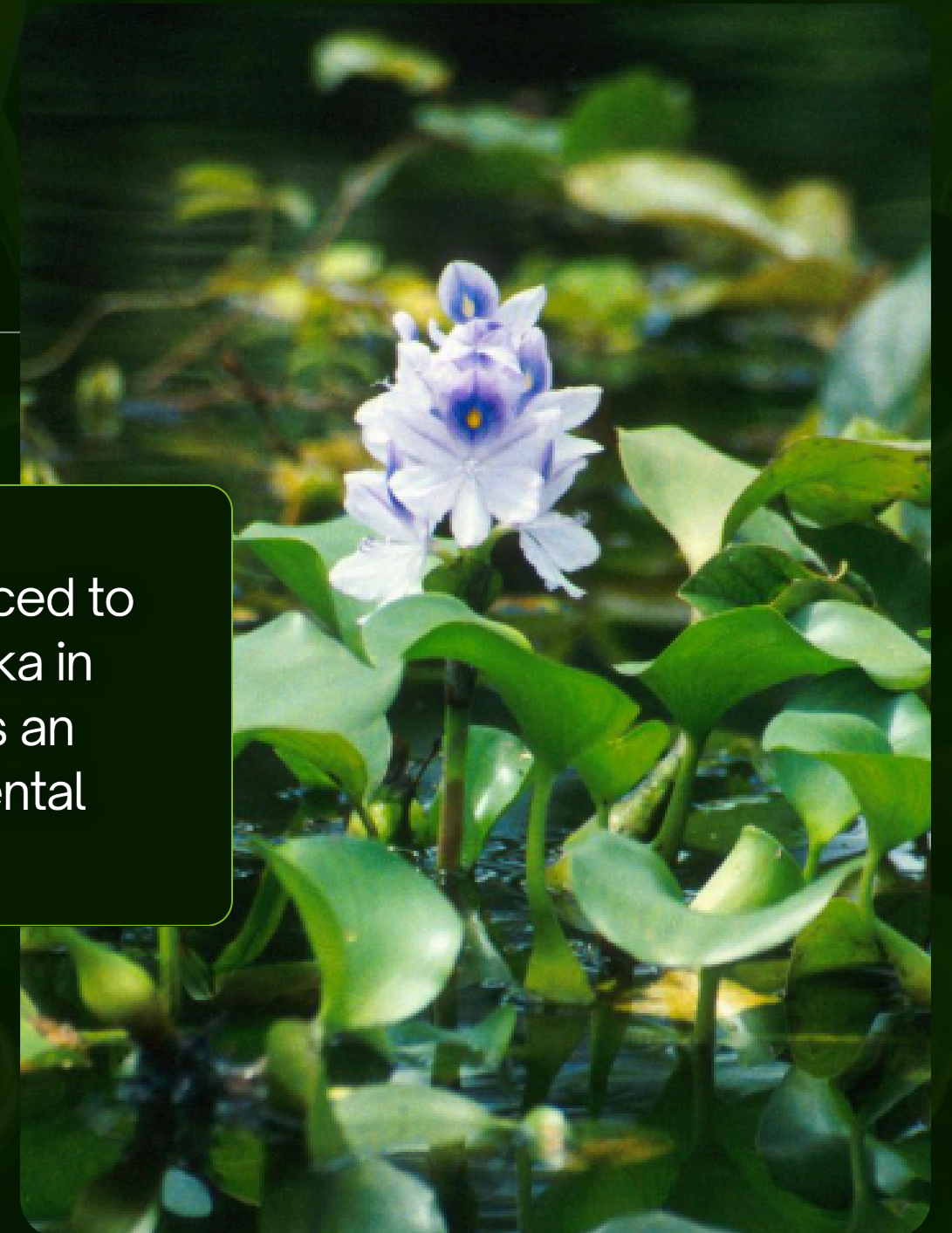
Presented By
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introduction

Water Hyacinth is an invasive aquatic plant that spreads rapidly and harm the aquatic biodiversity.

- Block sunlight and reduce oxygen percentage in water.
- Affects the fish populations other living organisms.
- Hinder fishing, irrigation and navigation on surface.
- Reduce the quality of water and other aquatic plants.

Introduced to Sri Lanka in 1905 as an ornamental plant.



Project Objectives

1

Automate
water Hyacinth
Removal
process

2

Detection
collection and
transportation
of plants

3

Self-Driving
robot
which avoids
other obstracles



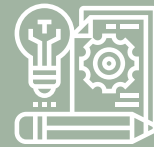
PROJECT ROADMAP



Problem
Definition



Literature
Review



Developing an
Model



Designing the
Prototype



Testing and
Implementation



Components Review

- | | | | |
|---|-----------------|----|--------------------------|
| 1 | Arduino UNO | 7 | Ultrasonic Sensor Module |
| 2 | Gear Motors | 8 | Push Buttons |
| 3 | Relay Module | 9 | Load Cell Module |
| 4 | Motor Driver | 10 | LCD Display |
| 5 | Servo Motor | 11 | Buzzer & LED Bulb |
| 6 | Buck Converters | 12 | Breadboard |



Autonomous process for detection, removal, collection and transportation of water hyacinth

Developed using cost-effective and reused materials

Portable structure which is flexible for the future enhancements

The Conclusion AquaHyel



Thank You!

Q & A